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1 Einführung

Wir betrachten das dynamische Transportproblem

$$\inf \int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dxdt + \int_0^1 \int_{\Omega} \frac{|f_t|^2}{2\rho_t} dxdt + \int_0^1 \int_{\Gamma} |\rho(t) - \rho_d|^2 dxdt$$

mit Kurve $\Gamma \subset (\Omega \times (0, 1))$ und Nebenbedingungen

$$\begin{aligned} \frac{\partial \rho_t}{\partial t} + \nabla \cdot (J_t + \rho_t v_t) &= f_t \quad \text{auf } \Omega \times (0, 1) \\ (J_t + \rho_t v_t) \cdot n &= 0 \quad \partial\Omega \times (0, 1). \end{aligned}$$

Dabei bezeichnet J_t das Energiefunktional zum Zeitpunkt t , f_t ist die Quelle, ρ_t ist ???, v_t das Vektorfeld und Datenterm. Mit Neumann-Randbedingungen haben wir halt ein abgeschlossenes System.

Um dieses Problem zu lösen, benötigt es einige Grundlagen aus Maßtheorie und Funktionalanalysis, eine Einführung in optimalen Transport und PDE/Variationsrechnung/Konvexe Analysis. Dann erarbeiten wir Existenzresultate und lösen das Problem noch numerisch.

Gegeben ein Vektorfeld und einer Datenkurve, welche auch eine Dichte im gewissen Sinne ist: Was war der Ursprung?

Bisherige Quellen: Santambrogio 4.1.2 (und OT-Theorie), Folland Maßtheorie + Funkana, Alt Funkana, OT von Figalli, Gradient Flows von Ambrosio/Gigli, Evans PDE, Benamou Brenier Paper, Variationsrechnung Skript Schmidt und Struwe

2 Mathematical Background

Das zu behandelnde Thema benötigt Kenntnisse und Grundlagen aus unterschiedlichsten mathematischen Teilgebieten. So geben wir einen Überblick über Maßtheorie, insbesondere Radonmaße und den Satz von Radon-Nikodym, grundlegende Definitionen aus der Funktionalanalysis - unterschiedliche Konvergenzarten (schwach und schwach-*) und in diesem Zug auch Dualräume. Darüber hinaus brauchen wir noch grundlegende Resultate aus der Funktionalanalysis und PDE-Theorie, etwa Lösungstheorie und Methoden der Variationsrechnung. Hier auch noch Konvexe Analysis?

2.1 Maßräume

2.1.1 Grundlegende Definitionen und Eigenschaften

Im Kontext von Transporttheorie, wo uns interessiert, wie wir möglichst effizient Masse von einem zum anderen Standort transportieren können, benötigen wir eine Idee von dem Konzept Masse und wie diese quantifiziert werden kann. Naheliegend finden wir dazu Antworten in der Maßtheorie. Wir beschäftigen uns erst mit dem Begriff eines Maßes und spezielle Klassen dieses Begriffs sowie zentrale Eigenschaften von Elementen, die wir messen können und überlegen uns, in welchem Sinne eine Ableitung für Maße existiert.

Definition 1 (σ -Algebra). Sei X eine Menge. Ein Mengensystem $\mathcal{A}$ über X heißt **σ -Algebra**, wenn die Eigenschaften

1. $\emptyset \in \mathcal{A}$,
2. $A \in \mathcal{A} \implies A^C \in \mathcal{A}$,
3. $(A_i)_{i \in I} \in \mathcal{A} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,

erfüllt sind.

Definition 2 (Signiertes Maß). Eine Abbildung $\mu : \mathcal{A} \rightarrow (-\infty, \infty]$ heißt **signiertes Maß**, wenn gilt

1. $\mu(\emptyset) = 0$,
2. $\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$

für paarweise disjunkte A_i . Zusätzlich nennen wir Abbildungen $\tilde{\mu} : \mathcal{A} \rightarrow [0, \infty]$ mit den gleichen Eigenschaften *positive Maße* und Abbildungen $\mathbb{P} : \mathcal{A} \rightarrow [0, 1]$ heißen *Wahrscheinlichkeitsmaße*.

Den Raum $(X, \mathcal{A})$ bezeichnen wir als **Messraum** und Elemente des Maßraums heißen *messbare Mengen*. Messbare Mengen E mit der Eigenschaft $\mu(E) = 0$ heißen *Nullmengen* bezüglich μ oder auch μ -Nullmengen. Ist es möglich, den Raum X als abzählbare Vereinigung von Mengen $A_1, \dots, A_n$ mit $\mu(A_i) < \infty$ für alle $i \in \mathbb{N}$ zu schreiben, so heißt das Maß σ -endlich. Ein Messraum ausgestattet mit einem Maß, also $(X, \mathcal{A}, \mu)$ heißt **Maßraum**.

Definition 3 (Erzeugte σ -Algebra). Sei $\mathcal{M} \subset \mathcal{P}(X)$. $\sigma(\mathcal{M}) = \bigcap_{\mathcal{A} \in \mathcal{F}(\mathcal{M})} \mathcal{A}$ heißt die *erzeugte σ -Algebra*, wobei $\mathcal{F}(\mathcal{M}) := \{\mathcal{M} \subseteq \mathcal{A} \mid \mathcal{A} \text{ ist } \sigma\text{-Algebra auf } X\}$.

Definition 4 (Borel- σ -Algebra). Sei $(X, \mathcal{O})$ ein topologischer Raum, wobei $\mathcal{O}$ das Mengensystem der offenen Mengen ist. $\sigma(\mathcal{O})$ ist die *Borelsche- σ -Algebra* und wir schreiben $\mathcal{B}(X) := \sigma(\mathcal{O})$.

Eine Klasse von besonders netten Maßen sind sogenannte Radon-Maße. Diese sind intuitiv, da kompakten Mengen ein endliches Maß zuordnen und sich Mengen auch approximieren lassen.

Definition 5 (Radon-Maße). Ein signiertes Maß $\mu : \mathcal{A} \rightarrow [-\infty, \infty)$ heißt Radon-Maß, wenn gilt

- $\mu(K) < \infty$ für $K \in \mathcal{A}$ kompakt
- Für jedes $E \in \mathcal{A}$ existiert ein $B \in \mathcal{B}(X)$ mit $B \supseteq E$ und $\mu(B) = \mu(E)$

TODO sigma-additiv, $X = \mathbb{R}^d$ definieren oder allgemeiner - wollen wir Polish Spaces oder brauchen wir bestimmte Eigenschaften?

Definition 6 (Variationsmaß). Für ein σ -additives signiertes Maß $\mu : \mathcal{A} \rightarrow [-\infty, \infty)$ definieren wir das Totalvariationsmaß

$$|\mu|(B) := \sup \left\{ \sum_{i=1}^{\infty} |\mu(B_i)| \mid \bigcup_{i=1}^{\infty} B_i = B \right\} \quad B \in \mathcal{B}(X).$$

Mit $\|\mu\|_{TV} = |\mu|(X)$ definieren wir die Totalvariations-Norm.

Prop? Lemma? Eig? Es gilt für alle $B \in \mathcal{B}(X)$

$$\begin{aligned} |\mu|(B) &= \inf\{|\mu|(U) \mid B \subset U, U \text{ offen}\} \\ |\mu|(B) &= \sup\{|\mu|(K) \mid B \supset K, K \text{ kompakt}\} \end{aligned}$$

TODO: Zeige, dass durch diese Norm ein Banachraum definiert ist – an späterer Stelle dann Dualraum zeigen. Hier auch noch vektorwertige Maße?

Für maßwertige Lösungen von PDE benötigen wir den zentralen Satz von Radon-Nikodym. Hierzu braucht es vorbereitende Definitionen

Definition 7 (Absolutstetigkeit von Maßen). Ein Maß μ heißt absolutstetig bezüglich ν mit Notation $\mu \ll \nu$, wenn für alle $A \subset \Omega$ mit $\nu(A) = 0$ auch $\mu(A) = 0$ gilt.

Zusätzlich haben wir mehr oder weniger das Gegenteil von Absolutstetigkeit.

TODO andere Definition verwenden?

Definition 8 (Singularität von Maßen). Seien μ, ν signierte Maße auf $(X, \mathcal{A})$. Wir nennen die zwei Maße singular, wenn Mengen $E, F \in \mathcal{A}$ existieren mit $E \cap F = \emptyset, E \cup F = X$ und $\mu(E) = 0, \nu(F) = 0$. Wir schreiben dabei $\mu \perp \nu$.

Theorem 2.1.1 (Hahn-Zerlegung). Sei μ ein signiertes Maß auf $(X, \mathcal{A})$, dann existieren Mengen P mit $\mu(P) \geq 0$ und N mit $\mu(N) \leq 0$ für die $P \cup N = X$ und $P \cap N = \emptyset$ gilt.

Beweis S 99

Theorem 2.1.2 (Jordan-Zerlegung). Sei μ ein signiertes Maß, dann existieren zwei positive Maße μ^+, μ^- , so dass $\mu = \mu^+ - \mu^-$ und $\mu^+ \perp \mu^-$.

Hier noch Beweis

Theorem 2.1.3 (Radon-Nikodym). Sei ν ein signiertes Maß und μ ein positives Maß, die jeweils σ -endlich auf $(X, \mathcal{A})$ sind. Es existieren eindeutige σ -endliche signierte Maße λ, ρ auf $(X, \mathcal{A})$ mit

$$\begin{aligned} \lambda &\perp \mu, \\ \rho &\ll \mu, \\ \nu &= \lambda + \rho. \end{aligned}$$

Zusätzlich existiert eine μ -integrierbare Funktion $f : X \rightarrow \mathbb{R}$ mit $d\rho = f d\mu$. f heißt dabei Radon-Nikodym-Ableitung und wird geschrieben als $\frac{d\nu}{d\mu} // d\nu = \frac{d\nu}{d\mu} d\mu$.

Es gilt $g \in L^1(\nu) \implies g(d\nu/d\mu) \in L^1(\mu)$ und $\int g d\nu = \int g \frac{d\nu}{d\mu} d\mu$.

Der Raum von Maßen ist konvex (wieso? Das ist super wichtig zu verstehen) Wegen Radon-Nikodym lässt sich jedes absolutstetige Maß schreiben als $\rho = f(x)dx$ mit $f \in L^1, \int f = 1$ (WMaß) und $L^1 \subset \mathcal{P}$ TODO. Vektor-wertige Maße?

2.2 Functional Analysis TODO

Nachdem das Problem das Konzept von Operatoren auf unendlichdimensionalen Räumen benötigt, führen wir zunächst grundlegende Begriffe wie Konzepte ein. Im Folgenden ist X dabei ein normierter $\mathbb{K}$ -Vektorraum. Ist X vollständig, so heißt er *Banachraum*. Enthält X eine abzählbare Teilmenge A mit $\overline{A} = X$, dann nennen wir X *separabel*. Ist zusätzlich auch Y ein normierter Vektorraum, so bezeichnen wir mit $L(X; Y)$ die Menge der stetigen linearen Abbildungen $T : X \rightarrow Y$, wobei wir T einen Operator nennen. Ein grundlegendes Resultat der Funktionalanalysis ist, dass lineare Operatoren genau dann stetig sind, wenn sie beschränkt sind. Mit X' bezeichnen wir den *Dualraum* von X . Elemente von X' sind dabei lineare Funktionale $\varphi : X \rightarrow \mathbb{K}$ (als Beispiel Integrale nennen). Dieses Objekt erlaubt es uns, zwei weitere Konvergenzarten zu definieren.

Wir nennen eine Folge $(x_n)_{n \in \mathbb{N}} \subset X$ *stark konvergent*, falls $\|x_n\|_X \xrightarrow{n \rightarrow \infty} \|x\|_X$.

Wir nennen eine Folge $(x_n)_{n \in \mathbb{N}} \subset X$ *schwach konvergenz*, falls für alle $\varphi \in X'$ gilt $\varphi(x_n) \xrightarrow{n \rightarrow \infty} \varphi(x)$.

Wir nennen eine Folge $(\varphi_n)_{n \in \mathbb{N}} \subset X'$ *schwach-* konvergent*, falls für alle $x \in X$ gilt $\varphi_n(x) \xrightarrow{n \rightarrow \infty} \varphi(x)$.

Hier interessiert uns die Konvergenz von (Wahrscheinlichkeits-)Maßen (TODO Motivation).

Theorem 2.2.1. *Banach-Alaoglu*

Theorem 2.2.2. *Riesz Representation*

Daraus dann folgern dass $C'_0(\Omega) = \mathcal{M}(\Omega)$ "Maße sind Lineare Funktionale im gewissen Sinne", braucht es für Duale Formulierung und Variationsprobleme

Reflexivität, Hahn-Banach,

Konvergenzarten von W-Maßen/Dichten im OT Kontext: narrow convergence, tightness, prokhorov, portmanteau? Generell: Was darf ich voraussetzen, was muss ich beweisen und was kann ich nur einmal erwähnen?

2.3 Variationsrechnung

Das vermutlich Verschieben nach 4: Warum brauchen wir das (Motivation), was sind die Methoden, die wir benötigen? Resultate verwenden für Lösung von Dualer Formulierung OT und DOT PDE-Theorie und damit L_p ? L_p Def, Hölder, Minkowski, Dualität; Sobolew, schwache Ableitung, Einbettung, Poincare; Distributionen, Testfunktionen, schwache Formulierung von PDE, Lax-Milgram

Variationsrechnung: Korezivität, Kompaktheit, lsc, Euler-Lagrange Gleichung und Lagrange Multiplikatoren (Direkte Methode)

3 Dynamical Optimal Transport

In this chapter we line out the basic framework in which our mathematical model is presented. We introduce the concept of optimal transport, first the static formulation of Monge-Kantorovich and in combination with the continuity equation the dynamical formulation that is then presented by the Benamou-Brenier formulation. The goal then is to present a result on the equivalence of solutions when comparing both formulations and also introduce a fleshed out version of the ALG2 algorithm presented in (BB), that is we solve the saddle point problem of the Benamou-Problem.

3.1 Optimal Transport Theorie

Monge, Kantorovich Speedrun

3.2 Dynamical Optimal Transport

We want to introduce the concept of dynamical optimal transport. In contrast to static optimal transport, we do not only care about finding an optimal map or optimal coupling to transport one distribution to another but are interested in a sense of flow in time, where for each time step $t \in (0, 1)$ the flow along a vector field is optimal. The necessary tools for that are a definition of a new metric, namely the Wasserstein metric, a new sense of derivative, namely a metric derivative and a little bit of convex analysis. We show that solutions of dynamic and static optimal transport are in fact equivalent and present a numerical method for solving the dynamical optimal transport problem, including an implementation.

While the total variation measure indeed takes into account the difference between two probability measures, it ignores a spacial distance, making it unfit for our purposes. A better choice is the Wasserstein-Distance

Definition 9 (Wasserstein Distance). Let X be a separable metric space, where each $\mu \in \mathcal{P}(X)$ is a radon measure and let $p \geq 1$. We define the p -th Wasserstein distance between two probability measures $\mu, \nu \in \mathcal{P}(X)$ as

$$W_p^p(\mu, \nu) := \inf \left\{ \int_{X^2} d(x, y)^p d\gamma(x, y) \mid \gamma \in \Gamma(\mu, \nu) \right\}$$

The concept of dynamic optimal transport is linked to the continuity equation, that can be interpreted as a condition to preserve mass during the flow of particles across a given velocity field and also allows us to then follow each single time step. Introducing the variable $\rho(t, x)$ with $t \in (0, 1)$ and $x \in \Omega$ is essentially the flow of our density in time-space with $\rho(0, x) = \mu(x)$ and $\rho(1, x) = \nu(x)$.
TODO schöner formulieren

The direction of a possible flow at time t and in x is given by a velocity field $v(t, x)$. We introduce lagrangian coordinates and denote by $X(t, x)$, $x \mapsto x_t$ the map of the position of some particle x to its position at time t , which can easily be applied to sets as well. The flow of each particle can be obtained by following the particle along the velocity field and we write

$$\frac{\partial X}{\partial t}(t, x) = v(t, X(t, x)), \quad X(0, x) = x$$

and when using sets specifically

$$\int_E \mu(x) dx = \int_{X(t, E)} \rho(t, x) dx \quad \forall t \in (0, 1).$$

The above formulation leads us to the realization that by change of variables we have

$$X(t, \cdot)_{\#} \mu = \rho(t, \cdot), \quad X(1, \cdot)_{\#} \mu = \nu.$$

Our goal is to rewrite the problem such that it merely depends on ρ, v and has the form

$$W_2^2(\mu, \nu) = \inf_{\rho, v} \int_0^1 \int_{\Omega} \rho(t, x) |v(t, x)|^2 dx dt$$

$$s.t. \begin{cases} \rho_t + \nabla \cdot (\rho v) = 0 \\ \rho(0, x) = \mu \\ \rho(1, x) = \nu \end{cases}$$

We begin with the small calculation

$$\begin{aligned} W_2^2(f, b) &\leq \int_{\Omega} \mu(x) |X(1, x) - x|^2 dx \\ &= \int_{\Omega} \mu(x) |X(1, x) - X(0, x)|^2 dx \\ &\stackrel{FTC}{=} \int_{\Omega} \mu(x) \left| \int_0^1 \frac{\partial X}{\partial t}(t, x) dt \right|^2 dx \\ &\stackrel{\Delta}{\leq} \int_{\Omega} \int_0^1 \mu(x) \left| \frac{\partial X}{\partial t}(t, x) \right|^2 dt dx \\ &= \int_0^1 \int_{\Omega} \mu(x) |v(t, X(t, x))|^2 dx dt \end{aligned}$$

This is a valid coupling since choosing $\gamma := (\text{id}, X(1, \cdot))_{\#} \mu$ leads to $\gamma \in \Gamma(\mu, \nu)$. To see that, we need μ and ν to be marginals of γ . Obviously, μ is a correct marginal of γ . Let $X(t, x)$ be the flow along the velocity field v and let ρ be such that it satisfies the continuity equation. We have

$$\rho(t, \cdot) = X(t, \cdot)_{\#} \mu \implies \nu = \rho(1, \cdot) = X(1, \cdot)_{\#} \mu$$

for the second marignal. Unnötig?

To evaluate the inner integral more easily, we define

$$S(x) = X(t, x), \quad S_{\#} \mu = \rho(t, x), \quad \varphi(y) = |v(t, y)|^2$$

and using change of variables yields

$$\begin{aligned} \int_{\Omega} \mu |v(t, X(t, x))|^2 dx &= \int_{\Omega} \mu \varphi(S(x)) dx \\ &= \int_{\Omega} \rho(t, y) \varphi(y) dy \\ &= \int_{\Omega} \rho(t, y) |v(t, y)|^2 dy \end{aligned}$$

and hence we have

$$W_2^2(\mu, \nu) \leq \int_0^1 \int_{\Omega} \rho(t, x) |v(t, x)|^2 dx dt$$

for all ρ, v that satisfy

$$\begin{cases} \rho_t + \nabla \cdot (\rho v) = 0, \\ \rho(0, x) = \mu, \\ \rho(1, x) = \nu. \end{cases}$$

Equality is obtained, if $X(1, x)$ is an optimal transport from μ to ν and $\frac{\partial X}{\partial t} = u(x)$ is constant in time. Assuming constant flow in time we simplify

$$X(t, x) = x + u(x)t \implies X(1, x) = x + u(x) =: T(x)$$

and solving for u gives us

$$\begin{aligned} u(x) = T(x) - x &\implies X(t, x) = x + (T(x) - x)t = x(1 - t) + tT(x) \\ &\implies v(t, X(t, x)) = u(x) = (T - I)(x). \end{aligned}$$

Therefore we found a formulation for an optimal transportation by

$$v(t, y) = (T - I)[(1 - t)I + tT]^{-1}(y)$$

and validated the problem formulation above. We can therefore write the optimization problem as

$$\begin{aligned} W_2^2(\mu, \nu) &= \inf_{\rho, v} \int_0^1 \int_{\Omega} \rho(t, x) |v(t, x)|^2 dx dt \\ s.t. \quad &\begin{cases} \rho_t + \nabla \cdot (\rho v) = 0, \\ \rho(0, x) = \mu, \\ \rho(1, x) = \nu. \end{cases} \end{aligned}$$

We try to solve the Lagrangian saddle-point problem. Note that by multiplying test functions $\varphi \in C_c^\infty((0, 1) \times \Omega)$ and partial integration, we obtain the weak form of the continuity equation

$$\int_{\Omega} (\nu \varphi(1, x) - \mu \varphi(0, x)) dx dt - \int_0^1 \int_{\Omega} (\rho \varphi_t + \rho v \cdot \nabla \varphi) dx dt$$

and hence the Lagrangian has the form

$$\mathcal{L}(\varphi, \rho, v) = \int_0^1 \int_{\Omega} \left(\frac{\rho |v|^2}{2} - \rho \varphi_t - \rho v \cdot \nabla \varphi \right) dx dt + \underbrace{\int_{\Omega} (\nu \varphi(1, x) - \mu \varphi(0, x)) dx}_{G(\varphi)}.$$

By defining $m := \rho v$ we have

$$\mathcal{L}(\varphi, \rho, m) = \int_0^1 \int_{\Omega} \left(\frac{|m|^2}{2\rho} - \rho \varphi_t - m \cdot \nabla \varphi \right) dx dt + G(\varphi).$$

3.3 Equivalence of the Static and Dynamic Formulation

The goal here is to show the relationship between static optimal transport, that is the Monge-Kantorovich formulation, and dynamical optimal transport. We will find that having a solution for one of these problems will then give us a solution to the other, making the problems equivalent. For this purpose, we define a few key terms.

Definition 10 (Metric Derivative). Let (X, d) be a metric space. We define the derivative of a curve $\gamma : [0, 1] \rightarrow X$ as

$$|\gamma'| (t) := \lim_{h \rightarrow 0} \frac{d(\gamma(t+h) - \gamma(t))}{|h|}.$$

Note that the absolute value of the denominator is necessary for the expression to be well-defined. For $t < s$ we get

$$d(c(t), c(s)) \leq \int_t^s |c'(x)| dx.$$

Also from the Rademacher-Theorem we conclude, that the metric derivative may exist for a.e. $t \in [0, 1]$ if the curve is Lipschitz-continuous. TODO?

Definition 11 (Absolutely Continuous Curves). A curve $c : [0, 1] \rightarrow X$ is said to be absolutely continuous, if there exists a $g \in L^1([0, 1])$ such that the inequality

$$d(c(t), c(s)) \leq \int_t^s g(x) dx$$

holds for all $t < s$. We write $c \in \text{AC}(X)$.

To show the first big result of this chapter, we need some auxiliary results.

Lemma 3.3.1. *If $\gamma \in \text{AC}(X)$, we can reparametrize the curve such that it is Lipschitz-continuous.*

Proof. Let $\gamma : [0, 1] \rightarrow (X, d)$ be absolutely continuous with $L(\gamma) = \int_0^1 |\gamma'(t)| dt < \infty$. The arc length is defined by $s(t) := \int_0^t |\gamma'(t)| dt$ with $s(0) = 0$ and $s(1) = L(\gamma)$. Reparametrizing by arc-length is defined by $\tilde{\gamma}(s) := \gamma(\tau(s))$, $s \in [0, T]$, $\tau(k) := \inf\{t \in [0, 1] \mid s(t) > k\}$. We compute

$$\begin{aligned} d(\tilde{\gamma}(a), \tilde{\gamma}(b)) &= d(\gamma(\tau(a)), \gamma(\tau(b))) \\ &\leq \int_{\tau(a)}^{\tau(b)} |\gamma'(r)| dr \\ &= s(\tau(b)) - s(\tau(a)) \leq b - a \end{aligned}$$

□

Lemma 3.3.2. *For $p > 1$ the optimal Transport map between two absolutely continuous measures exists.*

Proof. TODO

□

TODO hier noch Lemmata für das Theorem

One of the main theorems of this chapter is the following.

Theorem 3.3.3. *Let $(c_t)_{t \in [0, 1]} \subset \text{AC}(\mathcal{P}_p(X))$ be a family of absolutely continuous curves on the space of probability measures. For a.e. $t \in [0, 1]$ there exists a velocity field $v_t \in L^p(c_t, \mathbb{R}^d)$ with the properties*

- *the continuity equation is satisfied in the distributional sense*
- *we have $\|v_t\|_{L^p} \leq |c'(t)|$ for a.e. $t \in [0, 1]$.*

The idea of the proof is to construct a sequence of c_t and a corresponding sequence for v_t such that the continuity equation is solved, c_t^k is absolutely continuous for all k and the limit admits the desired inequality.

Proof. By Lemma ?? we assume μ to be Lipschitz. We define the sequence $\mu_t^k := \mu_t * \eta_k$, where ν_k is a Dirac sequence supported on $B(0, \frac{1}{k})$. This is a necessary step, since for example $\delta_x \in \mathcal{P}(\mathcal{X})$ is not absolutely continuous. Since

$$\text{supp}(\mu_t * \eta_k) \subset \Omega + B\left(0, \frac{1}{k}\right)$$

we extend the domain to a convex set $\Omega' \supset \Omega$. According to Lemma ??, for $t = \frac{i}{k}$, $i = 0, \dots, k$ the optimal transport map $T^{i,k} : \Omega' \rightarrow \Omega'$ from $\mu_{\frac{i}{k}}^k$ to $\mu_{\frac{i+1}{k}}^k$ exists as a consequence of absolute continuity. Defining the linear interpolation with constant speed

$$\mu_t^k := ((i+1 - tk)\text{id} + (kt - i)T^{i,k})_{\#} \mu_{\frac{i}{k}}^k$$

and deriving with respect to t yields the velocity

$$v^{i,k} := k(T^{i,k}(x) - x).$$

Since the continuity equation expects us not to have a particle indexed by its starting position but by the current position, we define

$$T_t^{i,k} := (i+1 - kt)\text{id} + (kt - i)T_k^{ik},$$

giving us the desired velocity field

$$v_t^k := v_{i,k}((T_t^{i,k})^{-1}).$$

For the interval $(\frac{i}{k}, \frac{i+1}{k})$, define

$$X_t(x) := T_t^{i,k}, \quad \mu_t^k = (X_t)_{\#} \mu_{\frac{i}{k}}^k$$

and by definition it holds

$$v_t^k(X_t(x)) = v_{i,k} = \dot{X}_t(x)$$

which defines an ordinary differential equations. Taking test functions and integrating by parts gives

$$\begin{aligned} \int_{\varphi} (y) d\mu_t^k(y) &= \int \varphi(X_t(x)) d\mu_{\frac{i}{k}}^k(x) \\ \iff \frac{d}{dt} \int \varphi(y) d\mu_t^k(y) &= \int \nabla \varphi(X_t(x)) \cdot v_t^k(X_t(x)) d\mu_{\frac{i}{k}}^k(x) \\ \implies \frac{d}{dt} \int \varphi d\mu_t^k &= \int \nabla \varphi(y) \cdot v_t^k(y) d\mu_t^k(y), \end{aligned}$$

which is exactly the weak form of the continuity equation, hence the pair (μ_t^k, v_t^k) solves $\partial_t \mu_t^k + \nabla \cdot (v_t^k \mu_t^k) = 0$ in the distributional sense.

To estimate an upper-bound of the newly obtained velocity field in order to extract a weak limit, we compute

$$\begin{aligned}
\|v_t^k\|_{L^p(\mu_t^k)}^p &= \int_{\Omega} |v^{i,k}|^p d\mu_{\frac{i}{k}}^k = k^p W_p^p \left(\mu_{\frac{i}{k}}^k, \mu_{\frac{i+1}{k}}^k \right) \\
&\leq k^p W_p^p \left(\mu_{\frac{i}{k}}, \mu_{\frac{i+1}{k}} \right) \\
&\leq k^p \left(\int_{\frac{i}{k}}^{\frac{i+1}{k}} |\mu'(s)| ds \right)^p \\
&\leq k \int_{\frac{i}{k}}^{\frac{i+1}{k}} |\mu'|^p(s) ds \leq \text{Lip}(\mu)
\end{aligned}$$

and

$$\int_a^b \|v_t^k\|_{L^p(\mu_t^k)}^p dt \leq \int_a^b |\mu'|^p(s) ds + \frac{2\text{Lip}(\mu)}{k}.$$

Define now vector measures E^k with test velocity field $\phi : \Omega' \times [0, 1] \rightarrow \mathbb{R}^d$ by

$$\int \phi(t, x) \cdot dE^k(t, x) := \int_0^1 \int_{\Omega'} \phi(t, x) \cdot v_t^k(x) d\mu_t^k(x) dt$$

and compute

$$\|E\| = \int_0^1 \|v_t^k\|_{L^1(\mu_t^k)} dt \leq \int_0^1 \|v_t^k\|_{L^p(\mu_t^k)} dt \leq \left(\int_0^1 \|v_t^k\|_{L^p(\mu_t^k)}^p dt \right)^{\frac{1}{p}} \leq C.$$

Since for $p > 1$ L^p is reflexive, E^k as a bounded sequence has a weakly converging subsequence. Also, by a triangle inequality argument, we can show that μ_t^k converges uniformly in time to μ_t in W_p . This gives us

$$\partial_t \mu_t + \nabla \cdot E = 0$$

in the distributional sense. REST TODO

□

As the goal is to show that a solution of the static problem admits one to the dynamic problem and vice-versa, we also want to show a converse result. TODO Lemma 5.21 states

Theorem 3.3.4. *If $(\mu_t)_{t \in [0,1]} \subset \mathcal{P}(\Omega)$ and for arbitrary t there exists $v_t \in L^p(\mu_t)$ satisfying $\int_0^1 \|v_t\|_{L^p(\mu_t)} dt < \infty$ and $\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0$, then $|\mu'|^p(t) \leq \|v_t\|_{L^p(\mu_t)}^p$ for a.e. t and $(\mu_t)_{t \in [0,1]}$ is absolutely continuous in $W_p(\Omega)$.*

Proof. Choosing a reparametrization as

$$s(t) := \int_0^t \|v_\tau\|_{L^p(\mu_\tau)} d\tau$$

for a given pair (μ, E) solving the continuity equation with $E_t = v_t \cdot \mu_t$ and $v_t \in L^p(\mu_t; \mathbb{R}^d)$, we get a uniform bound on $\|v_t\|_{L^p(\mu_t)}$ by and it also holds $\|E_t\| = \|v_t\|_{L^1(\mu_t)}$. Lemma ?? (5.21) guarantees the existence of a family $(\mu_t^\varepsilon)_t$ with a velocity field v_t^ε with the following properties:

1. μ^ε is Lipschitz in time and space, v^ε is Lipschitz in space and bounded, whilst being uniformly (bounded or continuous?) in time. $\mu^\varepsilon > 0$ everywhere.
2. We get μ_t^ε by following the flow of v^ε starting from μ_0^ε (see eq 5.5)
3. $\mu_t^\varepsilon \rightharpoonup \mu_t$ for all t and $E_t^\varepsilon \rightharpoonup E_t$ for a.e. t .
4. $\|v_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \mathcal{B}_p(\mu_t, E_t)$

DAS HIER INS LEMMA PACKEN UND HIER NUR STATEN

Taking a smooth function $\eta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\eta(t) = Ce^{-t}$ for $t \geq 1$ we construct $\eta^\varepsilon(z) = \varepsilon^{-d} \eta\left(\frac{|z|}{\varepsilon}\right)$, which lets us set

$$\mu_t^\varepsilon := \eta^\varepsilon * \mu_t, \quad E_t^\varepsilon := \eta^\varepsilon * E_t, \quad v_t^\varepsilon := \frac{E_t^\varepsilon}{\mu_t^\varepsilon}$$

With T_t being the flow generated by v^ε , we choose the transport plan between μ_t^ε and μ_{t+h}^ε as $\gamma = (T_t, T_{t+h})_\# \mu_0^\varepsilon \in \Gamma(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon)$. For the p -th Wasserstein-Distance, we compute

$$\begin{aligned} W_p^p(\mu_t^\varepsilon, \mu_{t+h}^\varepsilon) &\leq \int_{\Omega^2} |x - y|^p d\gamma(x, y) \\ &= \int_{\Omega} |T_t(x) - T_{t+h}(x)|^p d\mu_0^\varepsilon(x) \\ &\leq \int_{\Omega} \left(\int_t^{t+h} \left| \frac{d}{ds} T_s(x) \right| ds \right)^p d\mu_0^\varepsilon(x) \\ &\leq |h|^{\frac{p}{q}} \int_{\Omega} \int_t^{t+h} |v_s^\varepsilon(T_s(x))|^p ds d\mu_0^\varepsilon(x) \\ &= |h|^{\frac{p}{q}} \int_t^{t+h} \int_{\Omega} |v_s^\varepsilon(y)|^p d\mu_0^\varepsilon(y) ds \\ &\leq |h|^{\frac{p}{q}} \int_t^{t+h} \|v_s\|_{L^p(\mu_s)}^p ds. \end{aligned}$$

Then taking the limit is enough... TODO □

Corollary 3.3.5. *The vector field v_t in Theorem ?? (das erste) satisfies $\|v_t\|_{L^p(\mu_t)} = |\mu'| (t)$.*

3.4 Numerical Analysis of the Saddle-Point Problem

Recall the definition of the Lagrangian

$$\mathcal{L}(\varphi, \rho, m) = \int_0^1 \int_{\Omega} \left(\frac{|m|^2}{2\rho} - \rho \varphi_t - m \cdot \nabla \varphi \right) dx dt + G(\varphi).$$

We want to solve the saddle-point problem

$$\inf_{\rho, m} \sup_{\varphi} \mathcal{L}(\varphi, \rho, m)$$

with φ being the designated Lagrange-multiplier.

Lemma 3.4.1. We can rewrite $\frac{|m|^2}{2\rho}$ as the supremum of affine linear convex functions, namely

$$\sup_{(a,b) \in K} (a\rho + b \cdot m), \quad K := \left\{ (a,b) \in \mathbb{R} \times \mathbb{R}^n \mid a + \frac{|b|^2}{2} \leq 0 \right\}$$

Proof. Let (a^*, b^*) obtain the supremum of the set K . This is possible since K is obviously non-empty. Since $\rho \geq 0 \in \mathcal{P}(\Omega)$, a^* needs to be as large as possible. Rewriting the constraint of the set, we have

$$a^* = \frac{|b|^2}{2} \implies \sup_{(a,b) \in K} (a\rho + b \cdot m) = \sup_{b \in \mathbb{R}^n} \left(\frac{-\rho|b|^2}{2} + b \cdot m \right)$$

Since the last expression is concave, we have the maximum by deriving and setting to zero and as a consequence

$$b^* = \frac{m}{\rho} \implies \sup_{(a,b) \in K} (a\rho + b \cdot m) = \frac{|m|^2}{2\rho}.$$

□

As a consequence we rewrite the optimization problem as a linear one, that is

$$\inf_{\rho, m} \sup_{\varphi, (a,b) \in K} \int_0^1 \int_{\Omega} [(a - \varphi_t)\rho + (b - \nabla \varphi) \cdot m] dx dt$$

Benamou-Brenier introduced an augmented Lagrangian (TODO wieso eigentlich). Define

$$\mu := \begin{pmatrix} \rho \\ m \end{pmatrix}, \quad q := \begin{pmatrix} a \\ b \end{pmatrix}$$

then we can rewrite

$$\inf_{\rho, m} \sup_{\varphi} \mathcal{L}(\varphi, \rho, m) = \sup_{\mu} \inf_{\varphi, q} F(q) + G(\varphi) + \langle \mu, \nabla_{t,x} \varphi - q \rangle$$

where $G(\varphi)$ is defined as above and $F(q) = \begin{cases} 0, & q \in K \\ \infty & \text{else} \end{cases}$

TODO Eq 42 nochmal den Beweis abschreiben

Algorithmus: Erste Update in ϕ , dann q , dann μ .

TODO zu ϕ – was ist das überhaupt für ein Objekt?

Updating the algorithm in φ is finding a solution for the expression

$$\arg \min_{\varphi} \mathcal{L}_r(\varphi, q^{n-1}, \mu) = \arg \min_{\varphi} G(\varphi) + \langle \mu^n, \nabla \varphi - q^{n-1} \rangle + \frac{r}{2} \|\nabla \varphi - q^{n-1}\|^2 =: J(\varphi)$$

To avoid confusion, we denote by ψ the test-direction for differentiation. For the optimality condition we compute

$$\frac{d}{d\varepsilon} J(\varphi^n + \varepsilon \psi) \Big|_{\varepsilon=0} = 0$$

and get for each term

$$\begin{aligned}
G(\varphi^n + \varepsilon\psi) &= \int_{\Omega} [(\nu(\varphi(1, x) + \varepsilon\psi(1, x)) - \mu(\varphi(0, x) + \varepsilon\psi(0, x)))] dx \\
&= G(\varphi) + \varepsilon \int_{\Omega} (\nu\psi(1, x) - \mu\psi(0, x)) dx \\
&\implies \frac{d}{d\varepsilon} \Big|_{\varepsilon=0} G(\varphi^n + \varepsilon\psi) = G(\psi) \\
\frac{d}{d\varepsilon} \langle \mu^n, \nabla(\varphi + \varepsilon\psi) - q^{n-1} \rangle &= \frac{d}{d\varepsilon} \langle \mu^n, \nabla\varphi^n + \varepsilon\psi - q^{n-1} \rangle \\
&= \frac{d}{d\varepsilon} \langle \mu^n, \nabla\varphi^n - q^{n-1} \rangle + \varepsilon \langle \mu^n, \nabla\psi \rangle \\
&= \langle \mu^n, \nabla\psi \rangle \\
\frac{d}{d\varepsilon} \frac{r}{2} \|\nabla(\varphi^n + \varepsilon\psi) - q^{n-1}\|^2 &= r \langle \nabla\varphi^n - q^{n-1}, \nabla\psi \rangle
\end{aligned}$$

To solve for φ^n we need to integrate by parts. To make notation easier we write

$$\int_{\Omega} (\nu\psi(1, x) - \mu\psi(0, x)) + \int_{\Omega} (r(\nabla\varphi^n - q^{n-1}) + \mu^n) \cdot \nabla\psi = \int_{\Omega} (\nu\psi(1) - \mu\psi(0)) + \int_{\Omega} A \cdot \nabla\psi = 0$$

with $A = \begin{pmatrix} A_t \\ A_x \end{pmatrix}$. Then

$$\begin{aligned}
\int_{\Omega} A \cdot \nabla\psi &= - \int_{\Omega} (\nabla \cdot A)\psi + \int_{\Omega} A_t(1)\psi(1) - \int_{\Omega} A_t(0)\psi(0) \\
&\implies - \int_{\Omega} (\nabla \cdot A)\psi + \int_{\Omega} (\nu + A_t(1))\psi(1) - \int_{\Omega} (\mu + A_t(0))\psi(0) = 0
\end{aligned}$$

Since we assume periodic boundary conditions in space and Neumann boundary conditions in time, the boundary terms vanish and we calculate

$$\begin{aligned}
\nabla \cdot (r(\nabla\varphi^n - q^{n-1}) + \mu^n) &= 0 \\
\iff r\Delta\varphi^n - r\nabla \cdot q^{n-1} + \nabla \cdot \mu^n &= 0 \\
\iff \Delta\varphi^n = \nabla \cdot q^{n-1} - \frac{1}{r} \nabla \cdot \mu^n
\end{aligned}$$

TODO Jetzt noch Lösungstheorie: Direkte Methode, Sobolev Raum, H1 Skalarprodukt

For practical purposes, Benamou-Brenier proposed a slightly perturbed version with a small $\varepsilon > 0$ given by

$$-r\Delta\varphi^n + r\varepsilon\varphi^n = \nabla \cdot (\mu^n - rq^{n-1})$$

Zu q :

Äquivalenz zeigen. q is given by solving

$$\inf_{(a,b) \in K} \{(a - \alpha^n)^2 + |b - \beta|^2 \mid a + \frac{|b|^2}{2} \leq 0\}$$

where $p^n(t, x) = \begin{pmatrix} \alpha^n(t, x) \\ \beta^n(t, x) \end{pmatrix} = \nabla \varphi^n(t, x) + \frac{\mu^n(t, x)}{r}$. The Lagrangian of this convex problem leads us directly to an analytical solution. Define

$$\mathcal{L}(a, b, \lambda) = (a - \alpha)^2 + (b - \beta)^2 + \lambda \left(a + \frac{|b|^2}{2} \right)$$

then the KKT conditions

$$\begin{cases} \partial_a L = 0 \\ \partial_b L = 0 \\ \lambda \geq 0 \\ a + \frac{|b|^2}{2} \leq 0 \\ \lambda \left(a + \frac{|b|^2}{2} \right) = 0 \end{cases}$$

are satisfied. Stationarity leads us to

$$\begin{cases} \partial_a L = 2(a - \alpha) + \lambda \stackrel{!}{=} 0 \implies a = \alpha - \frac{\lambda}{2} \\ \partial_b L = 2(b - \beta)\lambda \stackrel{!}{=} 0 \implies b = \frac{2\beta}{2+\lambda} \end{cases}$$

In the case of $\lambda = 0$, then obviously $q^n = p^n$ (die zwei noch definieren). For $\lambda > 0$ we need to solve

$$F(\lambda) := \alpha - \frac{\lambda}{2} + \frac{2|b|^2}{2 \cdot (2 + \lambda)^2} = 0$$

This problem can be solved iteratively using Newton's method

$$\lambda^{(k+1)} = \lambda^{(k)} - \frac{F(\lambda^{(k)})}{F'(\lambda^{(k)})}$$

with $F'(\lambda^{(k)}) = -\frac{1}{2} - \frac{4\beta^2}{(2+\lambda)^3}$. Note that in the case $\lambda > 0$ the solution is unique, since the problem is strictly convex.

4 Data Fitting

For our modeling purposes we modify the constraints and rewrite the initial Benamou-Brenier formulation as

$$\inf_{(v_t, \rho_t)} \int_0^1 \int_{\Omega} \frac{|v_t|^2}{2\rho} dx dt$$

$$s.t. \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t v_t + \rho_t v_t^{wind}) = \alpha_{dif} \Delta \rho_t + f_t & \text{on } \Omega \times [0, 1] \\ (\rho_t v_t + \rho_t v_t^{wind} - \alpha_{dif} \nabla \rho_t) \cdot n = 0 & \text{on } \partial\Omega \times [0, 1] \\ \rho(0) = \rho_0, \quad \rho(1) = \rho_1 \end{cases}$$

Hier doch mu und nu verwenden für einheitliche Notation?

If we recall that

$$-\alpha_{dif} \Delta \rho_t = \nabla \cdot (-\alpha_{dif} \nabla \rho_t), \quad \nabla \rho_t = \rho_t \nabla \log \rho_t$$

we can reformulate the first constraint as

$$\begin{aligned} \partial_t \rho_t + \nabla \cdot (\rho_t v_t + \rho_t v_t^{wind}) - \alpha_{dif} \Delta \rho_t &= f_t \\ \iff \partial_t \rho_t + \nabla \cdot \underbrace{(\rho_t (v_t - \alpha_{dif} \nabla \log \rho_t))}_{=: J_t} + \rho_t v_t^{wind} &= f_t \end{aligned}$$

Using this notation and rewriting v_t gives us

$$\int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dx dt + \alpha_{dif} \int_0^1 \int_{\Omega} J_t \cdot \nabla \log \rho_t + \frac{\alpha_{dif}^2}{2} \int_0^1 \int_{\Omega} \underbrace{|\nabla \log \rho_t|^2}_{= \frac{|\nabla \rho_t|^2}{\rho_t}} \rho_t dx dt$$

and we simplyfy (dürfen wir das einfach so???) the whole problem as

$$\inf_{(\rho_t, J_t)} \int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dx dt + \alpha_{dif}^2 \int_0^1 \int_{\Omega} \frac{|\nabla \rho_t|^2}{\rho_t} \rho_t dx dt$$

$$s.t. \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t v_t + \rho_t v_t^{wind}) = f_t & \text{on } \Omega \times (0, 1) \\ (\rho_t v_t + \rho_t v_t^{wind}) \cdot n = 0 & \text{on } \partial\Omega \times (0, 1) \\ \rho_t(0) = \rho_0 \\ \rho_t(1) = \rho_1 \end{cases}$$

Adding two terms to complete the model, one for allowing creation of mass and one for fitting data for a given curve, we finally receive

$$\inf_{(f_t, J_t, \rho_t; \rho_0, \rho_1)} \int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dx dt + \alpha_f \int_0^1 \int_{\Omega} \frac{|f_t|^2}{2\rho_t} dx dt$$

$$+ \alpha_{dif}^2 \int_0^1 \int_{\Omega} \frac{|\nabla \rho_t|^2}{\rho_t} dx dt + \frac{\alpha_{in}}{2} \int_0^1 \int_{\Gamma} |\rho_t - \rho_{data}|^2 dx dt$$

$$s.t. \begin{cases} \partial_t \rho_t + \nabla \cdot (J_t + \rho_t v_t^{wind}) = f_t & \text{on } \Omega \times (0, 1) \\ (J_t + \rho_t v_t^{wind}) \cdot n = 0 & \text{on } \partial\Omega \times (0, 1) \end{cases}$$

and we may impose the condition on the initial and final distribution as above. Using the boundary condition we get for the first constraint

$$\begin{aligned}
& \int_0^1 \int_{\Omega} \partial_t \rho_t + \int_0^1 \int_{\Omega} \nabla \cdot (J_t + \rho_t v_t^{wind}) dx dt - \int_0^1 \int_{\Omega} f_t dx dt \\
&= \int_0^1 \frac{d}{dt} \left(\int_{\Omega} \rho_t dx \right) dt - \int_0^1 \int_{\Omega} f_t dx dt \\
&= \int_{\Omega} \rho_1 dx - \int_{\Omega} \rho_0 dx - \int_0^1 \int_{\Omega} f_t dx dt = 0
\end{aligned}$$

Here, our goal is also to solve the saddle-point problem, for which we need to calculate the Lagrangian in the first place. Taking a smooth test function $\phi_t : [0, 1] \times \Omega$ we obtain

$$\begin{aligned}
\mathcal{L}(\rho_t, J_t, f_t, \phi_t, \lambda_t) &= \int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dx dt + \alpha_f \int_0^1 \int_{\Omega} \frac{|f_t|^2}{2\rho_t} dx dt \\
&+ \alpha_{dif}^2 \int_0^1 \int_{\Omega} \frac{|\nabla \rho_t|^2}{\rho_t} dx dt \\
&+ \frac{\alpha_{inv}}{2} \int_0^1 \int_{\Omega} |\rho_t - \rho_{data}|^2 dx dt \\
&- \int_0^1 \int_{\Omega} \rho_t \partial_t \phi_t + J_t \cdot \nabla \phi_t + \rho_t v_t^{wind} \cdot \nabla \phi_t + f_t \phi_t dx dt \\
&+ \int_{\Omega} [\phi_1 \rho_1 - \phi_0 \rho_0] dx \\
&+ \lambda_t \left(\int_{\Omega} \rho_1 dx - \int_{\Omega} \rho_0 dx - \int_0^1 \int_{\Omega} f_t dx dt \right)
\end{aligned}$$

and the saddle point problem then is

$$\inf_{(\rho_t, J_t, f_t)} \sup_{(\phi_t, \lambda_t)} \mathcal{L}(\rho_t, J_t, f_t, \phi_t, \lambda_t).$$

TODO ADMISIIBLE SET

For a primal-dual projection algorithm we introduce convex functions and their respective Legendre-Fenchel transform. We define

$$\Psi(u, v) := \begin{cases} \frac{|v|^2}{2u} & u > 0 \\ 0 & u = 0, v = 0 \\ +\infty & \text{else} \end{cases}$$

and the two convex functions

$$\begin{aligned}
F^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, \nabla \rho_t) &:= \Psi(\rho_t, J_t) + \alpha_f \Psi(\rho_t, f_t) + \alpha_{dif}^2 \Psi(\rho_t, \nabla \rho_t) \\
\bar{F}^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, \nabla \rho_t) &:= F^{\alpha_f}(\rho_t, J_t, f_t, \nabla \rho_t) + \frac{\alpha_{inv}}{2} |\rho_t - \rho_{data}|^2
\end{aligned}$$

We recall that for a convex function F the Legendre-Fenchel transform has the form

$$F(x) = \sup_{x^*} \{ \langle x, x^* \rangle - F^*(x^*) \},$$

where F^* is the convex conjugate of F defined by

$$F^*(x^*) = \sup_x \{\langle x, x^* \rangle - F(x)\}.$$

Using this principle, we have

$$F^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, \nabla \rho_t) = \sup_{\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*} \{ \langle (\rho_t, J_t, f_t, \nabla \rho_t), (\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*) \rangle - (F^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*) \}$$

$$\bar{F}^{\alpha_f, \alpha_{dif}}(\rho_t, J_t, f_t, \nabla \rho_t) = \sup_{\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*} \{ \langle (\rho_t, J_t, f_t, \nabla \rho_t), (\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*) \rangle - (\bar{F}^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*) \},$$

where $F^{\alpha_f, \alpha_{dif}}$ is defined on $((0, 1) \times \Omega) \setminus \Gamma$ and $\bar{F}^{\alpha_f, \alpha_{dif}}$ on Γ . We can evaluate the conjugate by the indicator function, that is

$$\iota_C(x) := \begin{cases} 0 & x \in C \\ +\infty & \text{else} \end{cases}$$

by

$$(F^{\alpha_f, \alpha_{dif}})^*(\rho_t^*, J_t^*, f_t^*, \nabla \rho_t^*) = \iota_{\left\{ \rho_t^* + \frac{1}{2} \left(|J_t^*| + \frac{|f_t^*|}{\alpha_f} + \frac{|\nabla \rho_t^*|}{\alpha_{dif}^2} \right) \leq 0 \right\}}.$$

The given set is calculated similarly to what we did for the classical Benamou-Brenier formulation. For fixed ρ_t , we maximize

$$F^* = \sup_{\rho_t, J_t, f_t, \nabla \rho_t} \left\{ \rho_t \rho_t^* + J_t \cdot J_t^* + f_t f_t^* + \nabla \rho_t \nabla \rho_t^* - \frac{|J_t|^2}{2\rho_t} - \alpha_f \frac{|f_t|^2}{2\rho_t} - \alpha_{dif}^2 \frac{|\nabla \rho_t|^2}{2\rho_t} \right\}$$

seperately in every variable and obtain the set by calculating

$$\begin{cases} \sup_{J_t} \left\{ J_t \cdot J_t^* - \frac{|J_t|^2}{2\rho_t} \right\} = \frac{\rho_t |J_t^*|^2}{2} \\ \sup_{f_t} \left\{ f_t f_t^* - \alpha_f \frac{|f_t|^2}{2\rho_t} \right\} = \frac{\rho_t |f_t^*|^2}{2\alpha_f} \\ \sup_{\nabla \rho_t} \left\{ \nabla \rho_t \cdot \nabla \rho_t^* - \alpha_{dif}^2 \frac{|\nabla \rho_t|^2}{2\rho_t} \right\} = \frac{\rho_t |\nabla \rho_t^*|^2}{2\alpha_{dif}^2} \end{cases}$$

und ab hier ??? Vorzeichen in G und r>0?

5 Numerische Ergebnisse

Erklärung von FEM, das Vorgehen der Lösung, Code erklären.